

IIT Jodhpur Qualifier - BS

Time: 2 Hours | Maximum Marks: 198

Question Distribution

- **VARC:** 24 questions (suggested time 40 min)
- **DILR:** 18 questions (suggested time 40 min)
- **QA:** 24 questions (suggested time 40 min)
- Total: 66 questions | MCQ 60 | TITA (non-MCQ) 6

Marking Scheme

- MCQ: +3 for a correct answer, -1 for a wrong answer.
- TITA / Non-MCQ: +3 for a correct answer, no negative marking.
- Unanswered questions: no marks awarded or deducted.

General Instructions

- Read each question carefully before attempting.
- For MCQs, choose the single most appropriate option (A-D, or A-E where shown).
- For TITA questions, type the final numerical answer.
- Total duration is 2 Hours; the suggested per-section times are shown above.
- Sections may be attempted in any order; manage your time across all sections.

This is the answer-marked verification copy (correct option in green, with solution) and mirrors the content of the upload CSV.

Section: VARC (suggested 40 min)

Instructions (VARC-RC1): study the shared passage below and answer its 4 question(s).

Before the middle of the nineteenth century, most towns kept their own local time, setting clocks by the moment the sun stood highest overhead. A traveller moving west by carriage might find each new town a few minutes behind the last, and nobody minded, because journeys were slow enough that small differences never mattered. Railways changed that arithmetic. A train that left one city at noon and reached another whose clocks also read noon had, on paper, taken no time at all. Timetables became puzzles, and collisions on single-track lines were a real danger when two crews disagreed about when a siding had to be cleared.

The response was not a government decree but a private one. Railway companies began publishing schedules based on a single reference clock, usually the time kept at their largest terminus, and asked stations along the line to follow it. Towns that resisted found themselves consulting two clocks: one on the church tower for daily life and one at the station for departures. In practice the station clock won, because it governed the arrival of mail, freight and visitors.

Only later did legislatures ratify what commerce had already settled, dividing territories into broad zones within which every clock agreed. The reform is often described as a triumph of science, and astronomers did supply the measurements, but the pressure came from the ledger rather than the observatory. What made local time intolerable was not that it was inaccurate; by the standard of the sun it was perfectly accurate. It was that accuracy of that kind could not be shared. A network needs its members to agree on a convention, and the value of the convention lies almost entirely in the agreement, not in the truth of the thing agreed. That is why the change, once begun, spread quickly: each town that joined made joining more attractive for the rest.

Q1. [VARC-RC1 · mcsc · diff 0.5]

Which of the following best captures the central point of the passage?

- A) Uniform clock time spread because travellers grew tired of resetting their watches in every town they entered.
 - B) Uniform clock time spread because networks needed a shared convention, not because local time was wrong. ✓
 - C) Uniform clock time spread because governments saw that public safety required identical clocks everywhere.
 - D) Uniform clock time spread because astronomers at last produced measurements accurate enough to trust.
- Answer: B)** Uniform clock time spread because networks needed a shared convention, not because local time was wrong.

Solution: The passage insists that local solar time was accurate yet unshareable, and that railway networks needed agreement, so the convention mattered more than the truth of any clock. The astronomers version is rejected explicitly, since the pressure came from the ledger; the safety and traveller versions name effects the passage treats as symptoms, not as the driving reason.

Q2. [VARC-RC1 · mcsc · diff 0.5]

The passage attributes all of the following to the arrival of railways EXCEPT:

- A) confusion in drawing up timetables that different lines could actually work with
- B) danger of accidents when two crews disagreed about when a track would be clear
- C) a legal rule obliging church towers to show the same hour as the station clock ✓
- D) schedules published from a single reference clock kept at a large terminus

Answer: C) a legal rule obliging church towers to show the same hour as the station clock

Solution: The passage says legislatures later ratified broad zones; it never mentions any law about church towers, which simply lost influence in practice. Timetable puzzles, the collision risk on single-track lines, and schedules based on the terminus clock are all stated directly as consequences of the railways.

Q3. [VARC-RC1 · mcsc · diff 0.5]

It can be inferred from the passage that a town's church clock lost its authority mainly because:

- A) astronomers had declared solar time an unscientific way of setting clocks
- B) residents faced penalties for keeping time that differed from the railway's
- C) the church clock was found to be measurably slower than the railway's own clock
- D) the station clock decided when mail, goods and visitors would actually arrive ✓

Answer: D) the station clock decided when mail, goods and visitors would actually arrive

Solution: The passage says the station clock won because it governed the arrival of mail, freight and visitors, so practical dependence rather than any ruling settled the matter. Nothing suggests the church clock was slow, that penalties existed, or that astronomers condemned solar time; indeed solar time is called perfectly accurate.

Q4. [VARC-RC1 · mcsc · diff 0.5]

The author calls local time 'perfectly accurate' in order to:

- A) suggest that the towns were right to resist the schedules imposed on them
- B) concede that early railway schedules were built on faulty measurements
- C) praise the astronomers who first measured the sun's position from each town
- D) stress that its defect was not error but that it could not be shared ✓

Answer: D) stress that its defect was not error but that it could not be shared

Solution: The phrase sets up the next sentence, that accuracy of that kind could not be shared, so the concession about accuracy sharpens the argument about convention. The passage never calls railway measurements faulty, never endorses the resisting towns, and treats the astronomers as suppliers of data rather than as the point being made.

Instructions (VARC-RC2): study the shared passage below and answer its 4 question(s).

Beneath the floor of most forests lies a dense mesh of fungal threads, thinner than hair and far more numerous than roots. Many of these fungi wrap themselves around, or grow into, the roots of trees. The arrangement suits both sides: the fungus takes sugars that the tree makes in its leaves, and the tree gains water and minerals that the fungal threads, with their enormous surface area, can gather from soil far beyond a root's reach.

Because a single fungus can be linked to several trees at once, the threads also form a route between plants. Experiments in which one tree is fed carbon labelled so that it can be traced show that a small share of that carbon later turns up in neighbours connected to the same fungal partner. This finding has been retold in popular writing as evidence that forests behave like cooperatives, with vigorous trees deliberately feeding shaded seedlings.

The evidence supports something narrower. That a substance moves between two organisms does not establish that either one intended the transfer, or that the recipient benefited enough to matter. Carbon may simply follow a gradient, flowing from where it is abundant to where it is scarce, much as water does. The fungus, moreover, is not a neutral pipe. It is an organism with interests of its own, and it may retain much of what passes through, trading resources on terms that favour itself rather than any particular tree.

None of this makes the network unimportant. A seedling that gains even a modest supply of minerals in its first seasons may survive a drought it would otherwise not survive, and that is a large effect from a small transfer. The caution concerns the story we drape over the measurement. Describing the forest as generous is a metaphor; the data describe an exchange whose accounting has not yet been done.

Q5. [VARC-RC2 · mcsc · diff 0.5]

Which of the following best states the main argument of the passage?

- A) Fungal links between trees are real, but claims that trees deliberately share food outrun the data. ✓
- B) Fungal links between trees mostly injure the trees, because the fungus keeps nearly every resource.
- C) Fungal links between trees are so faint that the transfers recorded are probably measurement error.
- D) Fungal links between trees prove that a forest works as a cooperative rather than as a competition.

Answer: A) Fungal links between trees are real, but claims that trees deliberately share food outrun the data.

Solution: The passage accepts that carbon moves along fungal threads and that the effect on seedlings can matter, while rejecting the added story of deliberate sharing, so the caution is about interpretation rather than about the measurement. Calling the transfers error contradicts the drought example; the cooperative reading is the one being criticised; and the fungus is said to retain much, not nearly all.

Q6. [VARC-RC2 · mcsc · diff 0.5]

According to the passage, the fungal threads do all of the following EXCEPT:

- A) gather water and minerals from soil that lies beyond the reach of the roots
- B) produce sugars for themselves rather than take them from the tree's leaves ✓
- C) hold back for themselves a share of the resources that move along them
- D) connect one tree to another when a single fungus is joined to both of them

Answer: B) produce sugars for themselves rather than take them from the tree's leaves

Solution: The passage states that the fungus takes sugars the tree makes in its leaves, so making its own sugars instead is exactly what it does not do. Collecting water and minerals from distant soil, linking several trees through one fungus, and retaining part of what passes through are each stated in the text.

Q7. [VARC-RC2 · mcsc · diff 0.5]

It can be inferred that the author regards the carbon-tracing experiments as:

- A) flawed in method and therefore in need of repetition by other research teams
- B) sound in what they measure but weak support for the claims built on them ✓
- C) beside the point, since the quantities involved cannot affect survival
- D) conclusive proof that mature trees support shaded seedlings on purpose

Answer: B) sound in what they measure but weak support for the claims built on them

Solution: The author never questions the tracing itself; the objection is that movement of a substance does not establish intention or benefit, which makes the results reliable but thinner than advertised. Nothing calls the method flawed, the deliberate-feeding reading is the one resisted, and the drought example shows small transfers can matter.

Q8. [VARC-RC2 · mcsc · diff 0.5]

The comparison with water flowing down a gradient serves mainly to:

- A) show that the fungal threads carry mostly water rather than dissolved sugar
- B) argue that carbon moves too slowly to reach a neighbouring tree in a season
- C) suggest that the labelled carbon was washed through the soil by heavy rain
- D) offer a mechanism for the transfer that needs no intention by either tree ✓

Answer: D) offer a mechanism for the transfer that needs no intention by either tree

Solution: The gradient image supplies a plain physical reason for carbon to travel from abundance to scarcity, which removes the need to credit any tree with a purpose. The passage does not claim the threads carry mainly water, does not raise the speed of movement, and does not attribute the traced carbon to rainfall.

Instructions (VARC-RC3): study the shared passage below and answer its 4 question(s).

For much of the twentieth century, planners in growing cities treated the bicycle as a relic. Streets were widened, parking was added, and the space left for anyone on two wheels shrank to a painted strip beside fast traffic. Cycling did not disappear, but it became a choice for the brave.

In the last three decades many of those same cities have reversed course. The reason is mostly arithmetic. A lane of road carries far more people per hour on bicycles than in cars, because a bicycle needs about a tenth of the space and, when parked, almost none. Cities that could not afford to widen a single street found they could move more residents simply by changing what the existing width was used for.

Comfort turned out to matter more than exhortation. Where a city merely painted lines on the asphalt, ridership rose a little. Where it separated the cycle lane from moving traffic with a kerb or a row of parked cars, ridership rose sharply, and the new riders included many more children and older people. Surveys give a consistent explanation: most people are willing to cycle but unwilling to cycle beside lorries.

The change has limits. Bicycles work best for trips under about eight kilometres, so a city spread thinly over a wide area will always need other options, and hills and heavy rain reduce the numbers. Freight and emergency vehicles still need road space. What the experience of recent decades suggests is narrower than a slogan: for short journeys in dense districts, the cheapest way to add capacity is often to give a little room to bicycles, and the room only works if it feels safe.

Q9. [VARC-RC3 · mcsc · diff 0]

Which of the following best expresses the main idea of the passage?

- A) Cities have returned to cycling because residents demanded a healthier way of travelling each day.
- B) Cities have returned to cycling because protected lanes move many people cheaply on short trips. ✓
- C) Cities have returned to cycling because bicycles can replace cars on journeys of almost any length.
- D) Cities have returned to cycling because widening roads for cars proved technically impossible to do.

Answer: B) Cities have returned to cycling because protected lanes move many people cheaply on short trips.

Solution: The passage explains the reversal by space efficiency plus the sharp rise in ridership where lanes are separated from traffic, which is exactly capacity gained cheaply for short trips. Health is never mentioned; widening is described as unaffordable rather than impossible; and the text limits bicycles to trips under about eight kilometres.

Q10. [VARC-RC3 · mcsc · diff 0]

The passage mentions all of the following as limits on cycling EXCEPT:

- A) sloping ground and spells of heavy rainfall
- B) the price of buying and keeping up a bicycle ✓
- C) space that freight and emergency vehicles need
- D) journeys longer than roughly eight kilometres

Answer: B) the price of buying and keeping up a bicycle

Solution: Cost of ownership never appears in the passage, which discusses road space rather than household budgets. The final paragraph names distance beyond about eight kilometres, hills and heavy rain, and the road space still required by freight and emergency vehicles.

Q11. [VARC-RC3 · mcsc · diff 0.5]

It can be inferred that painted cycle lanes attract fewer new riders because:

- A) they are built only in districts where very few residents own a bicycle
- B) they are usually narrower than the width a bicycle actually needs to use
- C) they leave riders beside moving traffic, which most people will not take ✓
- D) they are marked in colours that drivers find difficult to notice in time

Answer: C) they leave riders beside moving traffic, which most people will not take

Solution: Paint leaves the rider next to moving vehicles, and the surveys quoted say most people are willing to cycle but unwilling to cycle beside lorries, so the missing element is separation. Lane width, paint colour and the location of the lanes are never discussed in the passage.

Q12. [VARC-RC3 · mcsc · diff 0.5]

The tone of the final paragraph is best described as:

- A) enthusiastic, urging cities to replace their car traffic with bicycles entirely
- B) sceptical, doubting that the recent changes to city streets achieved anything
- C) measured, marking out where cycle lanes help instead of stating a general rule ✓
- D) detached, refusing to take any position at all on the question under review

Answer: C) measured, marking out where cycle lanes help instead of stating a general rule

Solution: The paragraph lists limits and then states a deliberately narrow claim, calling it narrower than a slogan, which is qualified endorsement rather than advocacy. Enthusiasm is ruled out by the limits, scepticism by the endorsement, and detachment by the fact that a clear recommendation is offered.

Q13. [VARC-ODD-1 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) Many plants depend on animals to carry their seeds well away from the parent tree.
- (2) Fleshy fruits attract birds and mammals, which eat the pulp and drop the seeds elsewhere.
- (3) Other seeds carry small hooks that catch on fur and are brushed off much later.
- (4) Wind speed over open grassland is recorded by instruments mounted on tall poles.
- (5) Such journeys reduce the competition between a seedling and the plant that produced it.

Which sentence is the odd one out?

- A) (4) ✓
- B) (2)
- C) (5)
- D) (1)
- E) (3)

Answer: A) (4)

Solution: Sentences (1), (2), (3) and (5) form one chain: plants need animals to move seeds, two mechanisms for doing so (fruit pulp and hooks), and the benefit of the resulting distance. Sentence (4) is about instruments for measuring wind speed; it introduces neither seeds nor dispersal, so it is the misfit. (1) is the natural topic sentence and (5) the conclusion, so neither can be dropped without leaving the paragraph headless or unfinished; (2) and (3) are the paired examples that (1) demands.

Q14. [VARC-ODD-2 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) Before railways, every town kept its own clock, set roughly by the position of the sun.
- (2) Athletes often train at altitude in the hope of raising the oxygen capacity of their blood.
- (3) Timetables grew confusing once a train could cross many such local times in one day.
- (4) Railway companies therefore imposed a single standard time along the whole of their lines.
- (5) Governments later turned these private arrangements into the time zones still used today.

Which sentence is the odd one out?

- A) (5)
- B) (2) ✓
- C) (4)
- D) (1)
- E) (3)

Answer: B) (2)

Solution: The coherent paragraph runs (1), (3), (4), (5): local sun time, the problem trains created, the railways' fix, and its later official form. Sentence (2), about altitude training and blood oxygen, belongs to a different subject entirely and is the misfit. (3) supplies the problem that (4) solves, and (5) closes the sequence, so removing any of these breaks the causal chain instead of tightening it.

Q15. [VARC-ODD-3 · mcsc · diff 0.5]

Four of the following five sentences, when arranged properly, form a coherent paragraph on a single theme; one sentence does not belong.

- (1) Trees planted along city streets do far more than soften the look of concrete.
- (2) Their canopies shade the pavement and bring summer temperatures down noticeably.
- (3) Their roots take up rainwater that would otherwise overwhelm drains during a storm.
- (4) Their leaves also trap much of the dust that passing traffic throws into the air.
- (5) Architects still disagree about whether glass facades suit historic city centres.

Which sentence is the odd one out?

- A) (2)
- B) (4)
- C) (5) ✓
- D) (1)
- E) (3)

Answer: C) (5)

Solution: Sentences (1) to (4) form a unit: a claim that street trees are useful, then three uses (shade, drainage, dust capture). Sentence (5) stays in the city but switches to a dispute about building facades, mentioning no trees and no benefit, so it is the misfit. The three example sentences all begin from the tree itself and cannot be dropped without leaving (1) unsupported.

Q16. [VARC-JUMBLE-1 · mcsc · diff 0.5]

The four sentences below, when arranged in the correct order, form a coherent paragraph.

- (1) Riders who still feel unsafe on the road go on avoiding it for months afterwards.
- (2) Planners therefore judge such projects over a span of years rather than weeks.
- (3) When a city paints a cycle lane along a busy road, the effect is rarely immediate.
- (4) Only as the numbers slowly build does the lane start to feel ordinary and safe.

Select the correct order.

- A) 3412
- B) 1432
- C) 3142 ✓
- D) 1342

Answer: C) 3142

Solution: Sentence (3) is the only one that introduces the cycle lane as a new subject, so it opens. Its claim that the effect is not immediate is explained by (1), which says riders keep away for months. (4) then marks the turn, as numbers build the lane feels normal, and (2) draws the planning conclusion with 'therefore'. Orders beginning 1 fail because (1) starts with 'Riders who still feel unsafe on the road', which presupposes the lane already named in (3); 3412 puts the slow build before the avoidance it is supposed to follow.

Q17. [VARC-JUMBLE-2 · mcsc · diff 0.5]

The four sentences below, when arranged in the correct order, form a coherent paragraph.

- (1) After about a week it is lively enough to raise a loaf without any added yeast.
- (2) A sourdough starter is nothing more than flour and water left to attract wild yeasts.
- (3) Bakers who keep feeding it can go on using the same starter for decades.
- (4) Once bubbles appear, the mixture is fed every day with a little fresh flour.

Select the correct order.

- A) 2431
- B) 4231
- C) 2413 ✓
- D) 4213

Answer: C) 2413

Solution: Only (2) defines the starter, so it must come first; the pronoun 'the mixture' in (4) refers back to that flour and water. The sequence then follows the timeline: bubbles and daily feeding in (4), readiness after a week in (1), and continued use over decades in (3). Any order starting with (4) leaves 'the mixture' with no antecedent, and 2431 puts the decades-long habit before the starter is even usable.

Q18. [VARC-JUMBLE-3 · mcsc · diff 0.5]

The four sentences below, when arranged in the correct order, form a coherent paragraph.

- (1) Rising concern about the emissions of air travel then changed that calculation.
- (2) Cheap flights had made the sleeper look slow and expensive by comparison.
- (3) Several operators have since restored night routes between major European cities.
- (4) Night trains had almost disappeared from Europe by the early years of this century.

Select the correct order.

- A) 2413
- B) 4213 ✓
- C) 4231
- D) 2431

Answer: B) 4213

Solution: Sentence (4) states the situation to be explained, the near disappearance of night trains, and (2) gives its cause, cheap flights, while also depending on (4) for the term 'the sleeper'. (1) reverses the trend with 'then changed that calculation', and (3) reports the result with 'since'. Beginning with (2) leaves the comparison hanging, and placing (3) before (1) makes the revival precede the reason for it.

Q19. [VARC-JUMBLE-4 · mcsc · diff 0.5]

The four sentences below, when arranged in the correct order, form a coherent paragraph.

- (1) Grams simply remove the guesswork that volume measures quietly introduce.
- (2) Bakers who want repeatable results therefore reach for a kitchen scale instead.
- (3) Recipes in many countries still ask the cook to measure flour by the cupful.
- (4) The same cup, though, holds different weights depending on how tightly it is packed.

Select the correct order.

- A) 4321
- B) 3421 ✓
- C) 3241
- D) 4231

Answer: B) 3421

Solution: Sentence (3) introduces the cup as a measure, and (4) picks it up with 'The same cup, though' to state the problem, so this pair is fixed and must stand in that order. (2) draws the practical response with 'therefore', and (1) closes by explaining why weight settles the matter. Orders beginning with (4) leave 'the same cup' without a first mention, and 3241 places the scale before the flaw that motivates it.

Q20. [VARC-SUMMARY-1 · mcsc · diff 0]

Open-plan offices were introduced partly to encourage conversation between colleagues. In practice many workers find the constant background noise distracting and put on headphones in order to concentrate, which cuts off the very exchanges the layout was meant to create.

Which option best summarises the paragraph?

- A) Workers in open-plan offices wear headphones so that they can concentrate on their own tasks.
- B) Offices ought to be rebuilt with private rooms so that colleagues can talk to one another freely.
- C) Open-plan offices defeat their own aim, since noise drives workers to shut themselves off. ✓
- D) Quiet matters more to office productivity than the conversation between colleagues ever did.

Answer: C) Open-plan offices defeat their own aim, since noise drives workers to shut themselves off.

Solution: The paragraph has two halves, an aim (more conversation) and an outcome that undoes it (noise, headphones, no exchanges), and only the self-defeating-purpose statement captures both. Saying merely that workers wear headphones to concentrate reports one detail and drops the irony that is the point. The call to rebuild offices with private rooms is a recommendation the paragraph never makes. Ranking quiet above conversation distorts the text, which treats the loss of conversation as a cost, not a preference.

Q21. [VARC-SUMMARY-2 · mcsc · diff 0.5]

Delivery apps bring a restaurant orders it would never otherwise have received. They also take a share of every bill, and customers who once walked in now stay at home, so a busy kitchen can finish the month having sold more meals and kept less money.

Which option best summarises the paragraph?

- A) Restaurants that use delivery apps must pay a commission on every single order received.
- B) Customers order from home because delivery apps charge less than restaurants themselves do.
- C) Delivery apps lift a restaurant's order count while thinning the margin it earns on each. ✓
- D) Restaurants should abandon delivery apps and serve only customers who come in person.

Answer: C) Delivery apps lift a restaurant's order count while thinning the margin it earns on each.

Solution: The paragraph's shape is a gain in volume set against a loss in earnings, which is exactly what the statement about more orders but thinner margins preserves. The commission point is only one of the two forces named and ignores the lost walk-in trade, so it is too narrow. Telling restaurants to abandon the apps is advice the paragraph does not give. The claim that apps charge customers less is invented; the text says nothing about relative prices.

Q22. [VARC-SUMMARY-3 · mcsc · diff 0.5]

Where beavers return to a river they build dams, and parts of the valley flood. Farmers lose the use of some low-lying fields, but the new ponds slow floodwater heading downstream and hold water through dry spells, so the towns below gain what the fields lose.

Which option best summarises the paragraph?

- A) Beavers should be returned to every river, since their dams end both flooding and drought.
- B) Beaver dams cost nearby farmland while handing downstream towns a gain in water security. ✓
- C) Beavers build dams that flood low-lying fields and take useful farmland out of production.
- D) Farmers gain more from beaver ponds than the towns downstream do in long spells of dry weather.

Answer: B) Beaver dams cost nearby farmland while handing downstream towns a gain in water security.

Solution: The paragraph balances a loss borne by farmers against a benefit received downstream, so the statement naming both the farmland cost and the town's water gain is the only complete summary. Stopping at flooded fields records the cost and omits the benefit that the word 'but' introduces. Recommending beavers for every river, and claiming they end flooding and drought outright, goes far beyond 'slow' and 'hold'. Putting the farmers ahead of the towns reverses the direction of benefit the text states.

Q23. [VARC-SUMMARY-4 · mcsc · diff 0.5]

A translator of poetry cannot keep everything. Sound, rhythm and literal sense pull in different directions, and holding on to one of them usually means loosening the grip on another. Every finished translation is therefore a record of the choices its translator was willing to make.

Which option best summarises the paragraph?

- A) Skilled translators of poetry do manage to preserve sound, rhythm and literal meaning at once.
- B) Poetry is best read in the original, because any translation destroys what really matters in it.
- C) Translators of poetry find it hard to reproduce the rhythm of the original in another language.
- D) A poem's sound and sense compete, so translation records the trade-offs a translator accepts. ✓

Answer: D) A poem's sound and sense compete, so translation records the trade-offs a translator accepts.

Solution: The paragraph argues that the demands of a poem conflict and that the finished translation therefore shows what its translator chose to sacrifice, which the trade-off statement states directly. Naming rhythm alone as the difficulty covers one third of the tension and misses the conclusion about choices. The advice to read only originals adds a verdict the text avoids, since it never calls translation destructive. Claiming that skilled translators keep all three contradicts the premise that one must be loosened.

Q24. [VARC-SUMMARY-5 · mcsc · diff 0.5]

Cities that switch to bright white LED streetlights cut their electricity bills at once. Insects, however, are drawn to that part of the spectrum far more strongly than to the old orange lamps, and the moths gathering under the new lights are eaten or exhausted well before dawn.

Which option best summarises the paragraph?

- A) Moths keep away from white LED lights and so survive better than they did under orange ones.
- B) Cities ought to bring back orange streetlamps in order to protect their declining moths.
- C) The savings from white LED street lighting carry a hidden cost for night-flying insects. ✓
- D) White LED streetlights consume markedly less electricity than the orange lamps they replace.

Answer: C) The savings from white LED street lighting carry a hidden cost for night-flying insects.

Solution: The paragraph pairs an immediate financial benefit with a biological cost, and the statement about savings carrying a hidden cost to insects holds both halves together. Reporting only the lower consumption keeps the first sentence and discards the second. Urging a return to orange lamps is a policy proposal the text never advances. Saying moths keep away from the white lights inverts the paragraph, which has them gathering there and dying.

Section: DILR (suggested 40 min)

Instructions (DILR-1): study the shared data below and answer its 3 question(s).

Five runners, Farid, Lena, Mirko, Priya and Tomas, finished a 5 km club race in five different positions, from first to fifth. No two runners finished together.

- Priya finished ahead of Lena, and Lena finished ahead of Mirko. - Exactly one runner finished between Farid and Tomas, with Farid ahead of Tomas. - Priya did not finish first.

Q25. [DILR-1 · mcsc · diff 0.5]

Which runner finished third?

- A) Priya
- B) Tomas ✓
- C) Farid
- D) Lena

Answer: B) Tomas

Solution: The Farid-Tomas gap of exactly one runner allows the pairs 1-3, 2-4 or 3-5. Combining with Priya ahead of Lena ahead of Mirko and Priya not first leaves only Farid first, Priya second, Tomas third, Lena fourth, Mirko fifth. Priya is second, Lena fourth, and Farid first, so each of those choices misreads one clue.

Q26. [DILR-1 · mcsc · diff 0.5]

In which position did Lena finish?

- A) Fourth ✓
- B) Second
- C) Fifth
- D) Third

Answer: A) Fourth

Solution: The unique order is Farid, Priya, Tomas, Lena, Mirko, so Lena is fourth. Third belongs to Tomas and second to Priya; fifth comes from forgetting that Mirko must finish behind Lena.

Q27. [DILR-1 · mcsc · diff 0.5]

Which of the following statements is true?

- A) Lena finished immediately behind Tomas but ahead of Priya
- B) Tomas finished immediately ahead of Priya
- C) Mirko finished immediately behind Lena ✓
- D) Farid finished two places behind Priya

Answer: C) Mirko finished immediately behind Lena

Solution: In the order Farid, Priya, Tomas, Lena, Mirko, positions four and five are Lena and Mirko, so Mirko is immediately behind Lena. Tomas is behind Priya, not ahead of her; Farid is ahead of Priya rather than two places behind; and Lena finishes behind Priya, not ahead of her.

Instructions (DILR-2): study the shared data below and answer its 3 question(s).

Four members of a college club, Anya, Boris, Carla and Dev, earned points in a month-long fundraising drive. Together they earned 120 points.

- Boris earned 6 points more than Carla. - Anya earned exactly twice as many points as Dev. - Dev earned 16 points.

Q28. [DILR-2 · mcsc · diff 0.5]

How many points did Carla earn?

- A) 36
- B) 33 ✓
- C) 30
- D) 39

Answer: B) 33

Solution: Dev earned 16, so Anya earned 32 and the remaining 72 points are split between Boris and Carla with Boris 6 ahead: Carla 33 and Boris 39. The 30 figure comes from splitting 120 carelessly, 36 repeats Anya's total doubled wrongly, and 39 is Boris's total, not Carla's.

Q29. [DILR-2 · mcsc · diff 0.5]

Who earned the most points?

- A) Carla
- B) Dev
- C) Boris ✓
- D) Anya

Answer: C) Boris

Solution: The totals are Anya 32, Boris 39, Carla 33 and Dev 16, so Boris leads. Anya and Carla are close behind but neither tops 39, and Dev has the smallest total of all four members.

Q30. [DILR-2 · mcsc · diff 0.5]

By how many points does the highest total exceed the lowest total?

- A) 23 ✓
- B) 27
- C) 7
- D) 17

Answer: A) 23

Solution: Boris tops the list with 39 and Dev is lowest with 16, a gap of 23 points. The 7 comes from comparing Boris with Carla, 17 from using Anya as the top scorer, and 27 from subtracting Carla's total from Boris and Dev incorrectly.

Instructions (DILR-3): study the shared data below and answer its 4 question(s).

A beverage wholesaler reports last month's trade at its three depots. Table 1 gives the number of packs sold, in hundreds of packs. Table 2 gives the selling price of one pack, in euros, at that depot.

Table 1: Packs sold (in hundreds)

Category	Verona	Padua	Ferrara
Espresso blend	12	15	9
Herbal tea	8	6	10
Cocoa	5	9	11

Table 2: Price per pack (in euros)

Category	Verona	Padua	Ferrara
Espresso blend	20	20	25
Herbal tea	10	15	10
Cocoa	30	20	20

Q31. [DILR-3 · mcsc · diff 0.5]

How many hundreds of packs were sold in total at the Ferrara depot?

- A) 30 ✓
- B) 28
- C) 25
- D) 32

Answer: A) 30

Solution: The Ferrara column of Table 1 holds 9 packs of espresso blend, 10 of herbal tea and 11 of cocoa, so the depot sold $9 + 10 + 11 = 30$ hundred packs. The 25 figure is the Verona column total, 28 drops the cocoa row to 9, and 32 comes from reading the Padua espresso entry of 15 in place of 9.

Q32. [DILR-3 · mcsc · diff 0.5]

Which depot and category together generated the highest revenue?

- A) Padua, espresso blend ✓
- B) Ferrara, cocoa
- C) Verona, espresso blend
- D) Ferrara, espresso blend

Answer: A) Padua, espresso blend

Solution: Revenue for a cell is packs sold times price, in hundreds of euros: Padua espresso gives $15 \times 20 = 300$, Verona espresso $12 \times 20 = 240$, Ferrara espresso $9 \times 25 = 225$ and Ferrara cocoa $11 \times 20 = 220$. Padua espresso blend is therefore the largest at 30,000 euros; the Verona and Ferrara espresso entries trap candidates who compare only volumes or only prices.

Q33. [DILR-3 · mcsc · diff 0.5]

What was the combined revenue from cocoa across the three depots, in euros?

- A) 52,000
- B) 61,000
- C) 55,000 ✓
- D) 58,000

Answer: C) 55,000

Solution: Cocoa revenue in hundreds of euros is $5 \times 30 = 150$ at Verona, $9 \times 20 = 180$ at Padua and $11 \times 20 = 220$ at Ferrara, a total of 550 hundred euros, that is 55,000 euros. The 52,000 value comes from pricing Ferrara cocoa at 18 euros, 58,000 from swapping the Verona and Padua prices, and 61,000 from applying the 30 euro price to every depot.

Q34. [DILR-3 · mcsc · diff 0.5]

In how many of the nine depot and category combinations did revenue exceed 20,000 euros?

- A) 3
- B) 4 ✓
- C) 6
- D) 5

Answer: B) 4

Solution: The nine revenues in hundreds of euros are 240, 300, 225 for espresso blend, 80, 90, 100 for herbal tea and 150, 180, 220 for cocoa. Exceeding 20,000 euros means exceeding 200 hundred euros, which 240, 300, 225 and 220 do, giving 4 combinations. The count of 3 drops Ferrara cocoa at 220, while 5 and 6 wrongly admit the 180 and 150 cocoa figures.

Instructions (DILR-4): study the shared data below and answer its 3 question(s).

Four volunteers, Ines, Karim, Nadia and Omar, each work exactly one lab shift during a week, on Monday, Tuesday, Wednesday and Thursday, one volunteer per day. Each is also assigned a different single duty: filing, inventory, reception or training.

- Karim works on the day immediately after Nadia's shift.
- Ines works neither on Monday nor on Thursday.
- Omar's shift falls earlier in the week than Ines's shift.
- The volunteer working on Wednesday is assigned inventory.
- Ines is assigned reception, and Omar is assigned training.

Q35. [DILR-4 · mcsc · diff 0.5]

Which volunteer works on Monday?

- A) Omar ✓
- B) Karim
- C) Ines
- D) Nadia

Answer: A) Omar

Solution: Ines must work Tuesday or Wednesday, and Omar works before her. If Ines took Wednesday, the two days left for Nadia and Karim would not be consecutive, so Ines works Tuesday and Omar takes Monday, leaving Nadia on Wednesday and Karim on Thursday.

Q36. [DILR-4 · mcsc · diff 0.5]

Which volunteer and duty are correctly paired?

- A) Omar with reception
- B) Karim with inventory
- C) Nadia with filing
- D) Karim with filing ✓

Answer: D) Karim with filing

Solution: The schedule is Omar Monday, Ines Tuesday, Nadia Wednesday and Karim Thursday. Wednesday carries inventory, so that duty is Nadia's, while Ines has reception and Omar training; the only duty left for Karim is filing.

Q37. [DILR-4 · mcsc · diff 1]

Suppose Ines and Karim exchange shift days, every duty staying with the same person. Who then works on the day immediately before the inventory shift?

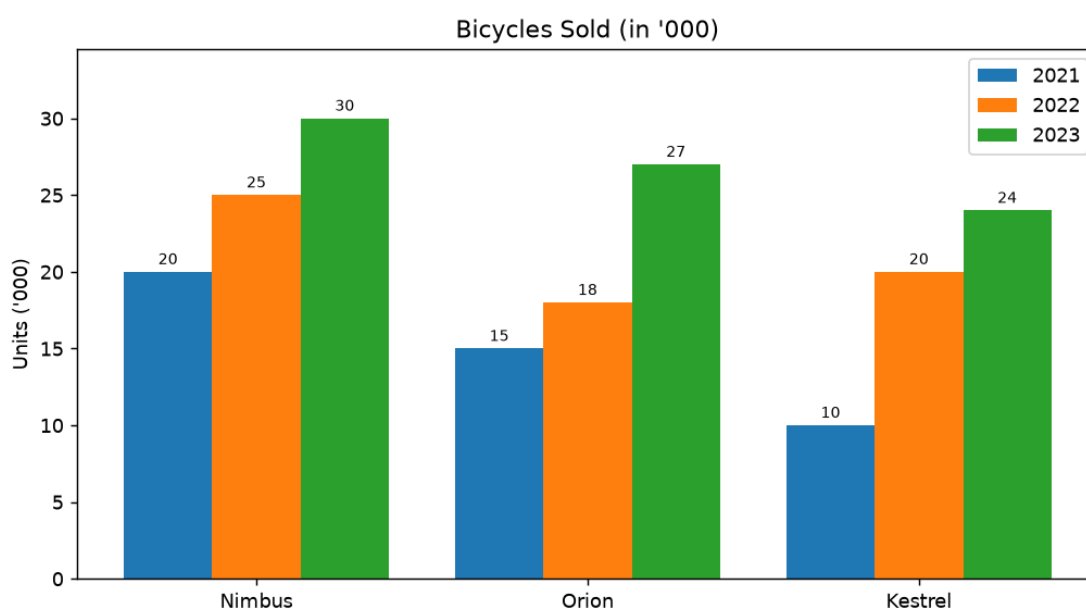
- A) Karim ✓
- B) Nadia
- C) Omar
- D) Ines

Answer: A) Karim

Solution: Originally Omar works Monday, Ines Tuesday, Nadia Wednesday and Karim Thursday, with inventory on Wednesday. After the exchange Karim works Tuesday and Ines Thursday, while Nadia keeps inventory on Wednesday, so Karim is the volunteer working the day before it. Ines has moved to Thursday, Omar stays on Monday, and Nadia holds the inventory shift itself.

Instructions (DILR-5): study the shared data below and answer its 5 question(s).

The bar chart shows the number of bicycles sold, in thousands, for three models of a manufacturer in each of the years 2021, 2022 and 2023.



Q38. [DILR-5 · mcsc · diff 0.5]

How many thousand bicycles were sold in total across the three models in 2022?

- A) 63 ✓
- B) 65
- C) 60
- D) 58

Answer: A) 63

Solution: For 2022 the bars read 25 thousand for Nimbus, 18 thousand for Orion and 20 thousand for Kestrel, so the total is $25 + 18 + 20 = 63$ thousand. The 58 figure keeps the 2021 Kestrel bar of 10 plus a misread, 60 rounds each bar to the nearest ten before adding, and 65 uses the 2021 Orion value of 15 with a doubled Kestrel reading.

Q39. [DILR-5 · mcsc · diff 0.5]

By what percentage did Kestrel sales grow from 2021 to 2023?

- A) 120%
- B) 140% ✓
- C) 58%
- D) 240%

Answer: B) 140%

Solution: Kestrel sales rise from 10 thousand in 2021 to 24 thousand in 2023, an increase of 14 thousand on a base of 10 thousand, which is 140%. The 240% value reports the 2023 figure as a percentage of the 2021 figure rather than the growth, 120% uses the 2022 value of 20 as the endpoint, and 58% divides the rise of 14 by the later year figure of 24 instead of the base.

Q40. [DILR-5 · mcsc · diff 0.5]

What is the ratio of total Orion sales over the three years to total Nimbus sales over the three years?

- A) 3 : 5
- B) 5 : 4
- C) 4 : 5 ✓
- D) 2 : 3

Answer: C) 4 : 5

Solution: Orion totals $15 + 18 + 27 = 60$ thousand and Nimbus totals $20 + 25 + 30 = 75$ thousand, so the ratio is 60 : 75, which reduces to 4 : 5. The 5 : 4 option reverses the two models, 2 : 3 comes from comparing only the 2021 bars of 15 and 20 wrongly reduced, and 3 : 5 arises from using the 2023 Orion bar against the Nimbus total.

Q41. [DILR-5 · mcsc · diff 0.5]

What were the average annual sales of Kestrel, in thousands, over the three years?

- A) 20
- B) 18 ✓
- C) 17
- D) 16

Answer: B) 18

Solution: Kestrel sells 10, 20 and 24 thousand units in the three years, a total of 54 thousand, so the average is 54 divided by 3, that is 18 thousand. The 16 value drops the 2023 bar to 18, 20 averages only the last two years, and 17 comes from taking the midpoint of the smallest and largest bars instead of the true mean.

Q42. [DILR-5 · mcsc · diff 0.5]

By how many thousand units did the three-model total in 2023 exceed the three-model total in 2021?

- A) 34
- B) 40
- C) 38
- D) 36 ✓

Answer: D) 36

Solution: The 2023 bars add to $30 + 27 + 24 = 81$ thousand and the 2021 bars add to $20 + 15 + 10 = 45$ thousand, so the excess is 81 minus 45, that is 36 thousand. The 34 value misreads the 2023 Orion bar as 25, 38 uses 2022 in place of 2021 for one model, and 40 comes from comparing 2023 against a rounded 2021 total of 41.

Section: QA (suggested 40 min)

Q43. [QA-MCQ-1 · mcsc · diff 0]

A school has 640 students, and 35% of them travel to school by bus. How many students travel by bus?

- A) 192
- B) 416
- C) 448
- D) 224 ✓

Answer: D) 224

Solution: 35% of 640 equals 640 multiplied by 35/100, which is 224 students, so 224 is correct. The value 416 counts the 65% who do not use the bus. The value 448 comes from taking 70% instead of 35%, and 192 comes from using 30% of 640.

Q44. [QA-MCQ-2 · mcsc · diff 0]

A trader buys a chair for 250 euros and sells it at a profit of 15% on the cost price. What is the selling price, in euros?

- A) 265.00
- B) 275.00
- C) 212.50
- D) 287.50 ✓

Answer: D) 287.50

Solution: The profit is 15% of 250, that is 37.50, so the selling price is 250 plus 37.50, giving 287.50 euros. The figure 212.50 applies a 15% loss instead of a profit, 265.00 adds a flat 15 euros rather than 15 percent, and 275.00 uses a 10% profit.

Q45. [QA-MCQ-3 · mcsc · diff 0]

A sum of 3300 euros is divided between Anna and Bruno in the ratio 4 : 7. How many euros does Bruno receive?

- A) 2100 ✓
- B) 1650
- C) 2200
- D) 1200

Answer: A) 2100

Solution: The 11 equal parts are worth $3300/11 = 300$ euros each, so Bruno's 7 parts come to 2100 euros. The value 1200 is Anna's 4 parts, 1650 splits the sum equally and ignores the ratio, and 2200 comes from using the fraction 2/3 instead of 7/11.

Q46. [QA-MCQ-4 · mcsc · diff 0]

Six readings taken during an experiment add up to 261. What is the average of the six readings?

- A) 37.3
- B) 52.2
- C) 44.5
- D) 43.5 ✓

Answer: D) 43.5

Solution: The average is the total divided by the count, so 261 divided by 6 gives 43.5. The value 52.2 divides by 5 readings, 37.3 divides by 7 readings, and 44.5 comes from rounding the total up to 267 before dividing.

Q47. [QA-MCQ-5 · mcsc · diff 0]

A circular flower bed has a diameter of 14 metres. What is the length of the fence needed to enclose it exactly once? (take $\pi = 22/7$)

- A) 22 m
- B) 154 m
- C) 88 m
- D) 44 m ✓

Answer: D) 44 m

Solution: The fence equals the circumference, which is π times the diameter, that is $(22/7)$ times 14 = 44 metres. The value 88 m treats 14 as the radius, 154 m is the area in square metres rather than a length, and 22 m halves the circumference.

Q48. [QA-MCQ-6 · mcsc · diff 0]

Two bells ring at intervals of 15 seconds and 18 seconds. If they ring together once, after how many seconds will they next ring together?

- A) 45
- B) 270
- C) 180
- D) 90 ✓

Answer: D) 90

Solution: They coincide after the LCM of 15 and 18, and since $15 = 3 \times 5$ and $18 = 2 \times 3 \times 3$, the LCM is $2 \times 3 \times 3 \times 5 = 90$ seconds. The value 270 is the plain product 15×18 without removing the common factor 3, 45 is only a multiple of 15, and 180 is twice the LCM.

Q49. [QA-MCQ-7 · mcsc · diff 0.5]

A sum of 12000 euros is lent at 8% per annum simple interest. What total amount, in euros, is repaid at the end of 2 years and 6 months?

- A) 14400 ✓
- B) 13920
- C) 14880
- D) 2400

Answer: A) 14400

Solution: The time is 2.5 years, so the interest is $12000 \times 8 \times 2.5 / 100 = 2400$ euros and the amount repaid is $12000 + 2400 = 14400$ euros. The value 13920 uses only 2 years, 14880 rounds the period up to 3 years, and 2400 is the interest alone without the principal.

Q50. [QA-MCQ-8 · mcsc · diff 0.5]

A cyclist rides from a village to a market 30 km away at 15 km/h and returns along the same road at 10 km/h. What is the average speed for the whole trip?

- A) 12.5 km/h
- B) 25 km/h
- C) 12 km/h ✓
- D) 11 km/h

Answer: C) 12 km/h

Solution: The trip takes $30/15 = 2$ hours out and $30/10 = 3$ hours back, so 60 km in 5 hours gives 12 km/h. The 12.5 km/h figure averages the two speeds directly, which is wrong because the times differ, 25 km/h adds the speeds, and 11 km/h comes from dividing 55 by 5.

Q51. [QA-MCQ-9 · mcsc · diff 0.5]

A water tank is $3/5$ full. After 24 litres are drawn out, the tank is $1/3$ full. What is the full capacity of the tank, in litres?

- A) 40
- B) 120
- C) 72
- D) 90 ✓

Answer: D) 90

Solution: The 24 litres correspond to $3/5$ minus $1/3$, that is $4/15$ of the capacity, so the capacity is 24 divided by $4/15 = 90$ litres. The value 40 comes from treating 24 litres as $3/5$ of the tank, 72 treats it as $1/3$ of the tank, and 120 comes from using a difference of $1/5$ instead of $4/15$.

Q52. [QA-MCQ-10 · mcsc · diff 0.5]

A sum of 12000 euros is lent for 2 years at 10% per annum. By how many euros does the compound interest, compounded annually, exceed the simple interest for the same period?

- A) 120 euros ✓
- B) 240 euros
- C) 252 euros
- D) 100 euros

Answer: A) 120 euros

Solution: Simple interest is $12000 \times 0.10 \times 2 = 2400$ euros. Compound interest is $12000 \times 1.1 \times 1.1 - 12000 = 2520$ euros, so the gap is 120 euros, which is simply the interest earned in year 2 on the first year's interest of 1200 euros. The 240 euro figure comes from doubling the gap by wrongly charging interest on interest twice, 252 euros is one tenth of the whole compound amount rather than the excess, and 100 euros ignores that the base for the extra interest is 1200 euros.

Q53. [QA-MCQ-11 · mcsc · diff 0.5]

Tap A alone can fill an empty tank in 15 minutes, while an outlet pipe B alone can empty a full tank in 20 minutes. If both are opened together when the tank is empty, how long will the tank take to fill completely?

- A) 60 minutes ✓
- B) 35 minutes
- C) 45 minutes
- D) 30 minutes

Answer: A) 60 minutes

Solution: In one minute tap A adds $1/15$ of the tank and pipe B removes $1/20$, so the net filling rate is $1/15 - 1/20 = 1/60$ of the tank per minute, giving 60 minutes. The 35 minute value adds the two times instead of combining the rates, 30 minutes doubles the smaller time by averaging the rates, and 45 minutes wrongly subtracts the times as $20 + 15 + 10$.

Q54. [QA-MCQ-12 · mcsc · diff 0.5]

A laboratory has 40 litres of a solution in which acid forms 15% of the volume. How many litres of pure water must be added so that acid forms exactly 10% of the resulting solution?

- A) 5 litres
- B) 10 litres
- C) 15 litres
- D) 20 litres ✓

Answer: D) 20 litres

Solution: The acid content stays fixed at $40 \times 0.15 = 6$ litres. For 6 litres to be 10% of the new volume the new volume must be 60 litres, so 20 litres of water is added. The 10 litre answer comes from treating the 5 percentage point drop as a 25% volume increase, 15 litres mixes up the concentration figure with a volume, and 5 litres assumes acid volume also changes.

Q55. [QA-MCQ-13 · mcsc · diff 0.5]

A box holds 5 red marbles and 3 green marbles. One marble is drawn at random, its colour is noted, and it is put back before a second marble is drawn. What is the probability that the two marbles drawn have the same colour?

- A) $17/32$ ✓
- B) $1/2$
- C) $13/28$
- D) $15/64$

Answer: A) $17/32$

Solution: With replacement each draw is independent, so $P(\text{both red}) = (5/8)(5/8) = 25/64$ and $P(\text{both green}) = (3/8)(3/8) = 9/64$. Adding gives $34/64 = 17/32$. The value $13/28$ is the answer for drawing without replacement, $15/64$ is the chance of one red and one green in a fixed order, and $1/2$ assumes the two colours are equally likely.

Q56. [QA-MCQ-14 · mcsc · diff 0.5]

A committee of 3 members is to be chosen from 5 men and 4 women. In how many ways can this be done if the committee must include at least one woman?

- A) 64
- B) 74 ✓
- C) 84
- D) 40

Answer: B) 74

Solution: Total selections of 3 from 9 people number 84, and those with no woman number 10, being the selections of 3 from the 5 men, so $84 - 10 = 74$ committees contain at least one woman. The 84 figure forgets the restriction, 64 subtracts the men themselves rather than the all-male triples, and 40 counts only one woman with two men.

Q57. [QA-MCQ-15 · mcsc · diff 0.5]

A closed solid cylinder has a base radius of 7 cm and a height of 10 cm. Taking $\pi = 22/7$, what is its total surface area?

- A) 594 square cm
- B) 440 square cm
- C) 902 square cm
- D) 748 square cm ✓

Answer: D) 748 square cm

Solution: Total surface area equals $2\pi r(h + r) = 2 \times (22/7) \times 7 \times 17 = 44 \times 17 = 748$ square cm. The 440 value is only the curved surface, 594 adds just one circular face instead of two, and 902 uses the volume-style factor by taking $h + 2r$ as the bracket.

Q58. [QA-MCQ-16 · mcsc · diff 1]

What is the smallest positive integer that leaves a remainder of 3 when divided by each of 6, 7 and 8, and is also exactly divisible by 11?

- A) 1683 ✓
- B) 1680
- C) 2523
- D) 171

Answer: A) 1683

Solution: Numbers leaving remainder 3 on division by 6, 7 and 8 have the form $168k + 3$, since the LCM of 6, 7 and 8 is 168. For divisibility by 11, $168k + 3$ leaves $3k + 3$ modulo 11, which is zero when $k = 10$, giving $1680 + 3 = 1683 = 11 \times 153$. The 1680 value drops the remainder, 171 is the first multiple of 11 in the sequence only if the modulus is mistaken as 84, and 2523 is the next valid number, not the smallest.

Q59. [QA-MCQ-17 · mcsc · diff 1]

Working in pairs, A and B finish a job in 12 days, B and C finish the same job in 15 days, and A and C finish it in 20 days. Working all three together, in how many days will they finish the job?

- A) 12 days
- B) 5 days
- C) 10 days ✓
- D) 16 days

Answer: C) 10 days

Solution: Adding the three pair rates gives $1/12 + 1/15 + 1/20 = 1/5$ of the job per day, but this counts every worker twice, so the combined rate of the three is $1/10$ per day and the time is 10 days. The 5 day answer forgets to halve the doubled rate, 12 days just repeats the fastest pair's time, and 16 days averages the three given times.

Q60. [QA-MCQ-18 · mcsc · diff 1]

In an arithmetic progression the sum of the first 15 terms is 600 and the 5th term is 22. What is the 20th term of this progression?

A) 112 ✓

B) 118

C) 100

D) 106

Answer: A) 112

Solution: The sum of 15 terms equals 15 times the 8th term, so the 8th term is 40. From the 5th term 22 to the 8th term 40 there are three common differences, so $d = 6$ and the first term is $22 - 24 = -2$. Hence the 20th term is $-2 + 19(6) = 112$. The 106 value stops at the 19th term, 118 goes one term too far, and 100 uses the 8th term as the first term.

Q61. [QA-TITA-1 · tita · diff 0]

In a geometric progression the first term is 5 and the common ratio is 3. What is the fourth term? Type your answer as an integer.

Answer (TITA): 135

Solution: Each term is three times the one before it, so the sequence runs 5, 15, 45, 135. Using the formula, the fourth term is $5 \times 3^3 = 5 \times 27 = 135$. Multiplying by the ratio four times instead of three gives 405, and adding 3 each time as if the progression were arithmetic gives 14.

Q62. [QA-TITA-2 · tita · diff 0]

What is the sum of all even numbers from 2 to 40, both included? Type your answer as an integer.

Answer (TITA): 420

Solution: The even numbers 2, 4, ..., 40 form 20 terms. Their sum is $2 \times (1 + 2 + \dots + 20) = 2 \times 210 = 420$. Equivalently, the average of the first term 2 and the last term 40 is 21, and 21×20 terms gives 420. Counting $40/2 = 20$ terms is the step most often missed; using 40 terms leads to the wrong total 840.

Q63. [QA-TITA-3 · tita · diff 0.5]

The population of a town is 12500. During the first year it rises by 20%, and during the second year it falls by 10%. What is the population at the end of the second year? Type your answer as an integer.

Answer (TITA): 13500

Solution: After the first year the population is $12500 \times 1.20 = 15000$. The 10% fall in the second year is applied to 15000, not to the original 12500, so the final figure is $15000 \times 0.90 = 13500$. Adding the two percentages into a single 10% net rise gives 13750, which is wrong because the second change uses the larger base 15000.

Q64. [QA-TITA-4 · tita · diff 0.5]

How many four-digit numbers can be formed using the digits 0, 1, 2, 3 and 4 if no digit is repeated in a number? Type your answer as an integer.

Answer (TITA): 96

Solution: The leading digit cannot be 0, so it has 4 choices from 1, 2, 3, 4. The remaining three places are filled from the four unused digits in $4 \times 3 \times 2 = 24$ ways. The count is $4 \times 24 = 96$. Ignoring the leading-zero restriction gives $5 \times 4 \times 3 \times 2 = 120$, and the 24 difference is exactly the strings beginning with 0.

Q65. [QA-TITA-5 · tita · diff 0.5]

A train 180 m long crosses a platform 240 m long in 21 seconds. What is the speed of the train in km/h? Type your answer as an integer.

Answer (TITA): 72

Solution: To clear the platform the train covers its own length plus the platform length, that is $180 + 240 = 420$ m, in 21 seconds. The speed is $420/21 = 20$ m/s, and multiplying by $18/5$ converts this to 72 km/h. Using only the 240 m platform gives $240/21$ m/s, about 41 km/h, and forgetting the $18/5$ conversion leaves the answer stuck at 20.

Q66. [QA-TITA-6 · tita · diff 1]

What is the smallest positive integer that leaves a remainder of 3 when divided by 5, a remainder of 4 when divided by 6, and a remainder of 5 when divided by 7? Type your answer as an integer.

Answer (TITA): 208

Solution: In each condition the remainder falls short of the divisor by exactly 2, since $5 - 3 = 6 - 4 = 7 - 5 = 2$. So the number plus 2 is divisible by 5, 6 and 7, whose LCM is 210. The smallest positive value is $210 - 2 = 208$, and a check confirms $208 = 41 \times 5 + 3$, $34 \times 6 + 4$, $29 \times 7 + 5$. Adding 2 instead of subtracting gives 212, and taking the LCM alone gives 210.